Threat-Aware Evaluation of Quantum Entanglement Routing and Qubit Allocation via Hybrid Contextual Bandits

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***Abstract*—Quantum entanglement routing requires joint path selection and qubit allocation under noisy, nonstationary, and adversarial conditions. Existing approaches often assume stationary links, fixed allocation rules, or offline optimization assumptions. When these assumptions fail, routing can degrade end-to-end entanglement quality and waste scarce quantum resources. To address these limitations, we introduce a threat-aware evaluation framework for stochastic, contextual/neural, adversarial, predictive, and hybrid bandit policies for joint quantum path selection and qubit allocation. Unlike previous work that fixes allocator policy and replay semantics, we vary threat regime, allocator policy, and replay capacity as first-class factors, enabling attribution of robustness to the bandit-policy–allocator–capacity interaction. Across thirteen bandit policies and five threat regimes, neural hybrids outperform non-contextual baselines by 18–24 percentage points (pp) in scenario-aggregated efficiency, sustain worst-case efficiency above 85% under stochastic threats, and remain more stable than adversarial-first EXP3-style designs under adaptive attacks. We also identify a capacity paradox: under `ThompsonSampling` allocation and T_b-type replay anchoring, mean OnlineAdaptive efficiency drops by 4.4 pp from $s{=}1$ to $s{=}1.5$, then recovers by 6.0 pp at $s{=}2$. Our cross-testbed evaluation on four external quantum-network testbeds confirms the main robustness trends while exposing scale- and topology-dependent limits. These results establish that context-aware neural policies paired with appropriate allocators provide deployment-grade robustness, while capacity scaling must be threat-matched to avoid predictability-induced collapse.**

***Index Terms*—Quantum networks, entanglement routing, multi-armed bandits, adversarial learning, qubit allocation, online optimization.**

I. Introduction

Quantum entanglement distribution is a core property for the quantum Internet, supporting quantum key distribution, distributed quantum computing, and sensing [1]–[3]. Reliable end-to-end entanglement is difficult because generation and swapping are probabilistic, quantum states are fragile, and decoherence makes path quality time-varying [4]–[6]. Unlike classical packet routing, quantum routing must establish and consume entanglement under limited memory coherence, probabilistic operations, and fidelity loss [6], [7]. These constraints tightly couple path selection with qubit allocation, since resource placement affects success probability and learner feedback [8]–[10].

Existing quantum-routing studies propose online path selection, benchmarking-driven routing, adaptive route selection, and adversarially robust learning [9]–[11], but they are often evaluated under different threat, topology, allocator, and replay assumptions [8], [12], [13]. This creates a matched-threat evaluation gap: it is unclear when contextual structure is necessary, when adversarial robustness dominates, and how allocator and capacity choices alter routing performance.

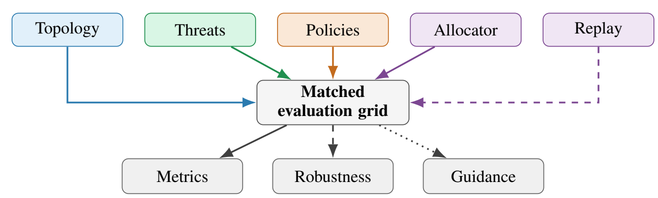


Fig. 1. Five matched inputs—topology, threat, policy, allocator, and replay—feed a shared evaluation grid to produce `Oracle`-normalized metrics, robustness comparisons, and deployment guidance. Matching all five inputs, simultaneously, exposes the `policy--allocator--capacity` interaction as the controlling robustness factor.

To address this lack of a unified and controlled evaluation framework for quantum-routing strategies, we introduce a threat-aware evaluation framework that compares stochastic/contextual, adversarial, predictive, and hybrid bandit policies for joint path selection and qubit allocation under matched threat, allocator, and replay-capacity settings. The evaluation pipeline is summarized in Figure 1. This evaluation provides practical guidance for designing quantum-network controllers that can maintain high entanglement efficiency and robustness under changing network conditions and adversarial disruptions. Across this controlled grid, pursuit–neural hybrids provide the strongest robustness–efficiency tradeoff, while replay capacity exhibits a threat-dependent capacity paradox: additional capacity can improve structured-disruption performance yet reduce robustness under adaptive attacks.

In summary, our work makes the following contributions:

- **Evaluation framework:** A threat-aware benchmark comparing bandit policies for joint path pick and qubit allocation across 8,241 matched threat–allocator–capacity tests.
- **Capacity paradox:** Capacity effects are threat-dependent: under `ThompsonSampling` allocation with base-horizon replay capacity (T_b), increasing replay-capacity scale from $s{=}1$ to $s{=}1.5$ causes a 4.4 percentage-point (pp) efficiency drop under reactive adaptive disruption, which then recovers by 6.0 pp at full scale ($s{=}2$). Allocator mismatch also lowers worst-case efficiency floors[1] by 15.6 pp, showing that replay capacity and allocation must be tuned together.
- **Deployment guidance:** Pursuit–neural hybrids sustain $\geq$85% worst-case efficiency while allocator choice induces

[1] Defined in Table I

10–15 pp swings, requiring routing policy and qubit allocation to be selected jointly.

- **Experimental findings:** Pursuit–neural hybrids achieve 87–96% efficiency, outperforming non-contextual baselines by 18–24 pp, and sustaining stability under strategic attacks.

The evaluation framework code and data are available at https://anonymous.4open.science/r/quantum_project-D2FF.

II. Related Work

Quantum routing and online path selection: Quantum networks distribute entanglement through repeaters and end nodes to support long-distance communication, distributed quantum computing, and sensing [1], [2]. Entanglement routing differs from classical packet routing because path quality depends on probabilistic entanglement generation and swapping, decoherence, fidelity loss, and limited qubit memory [4]–[7]. Prior quantum-routing work studies fidelity-aware path selection, multipath/inter-domain routing, benchmarking-based path estimation, resource-constrained routing, and adaptive or learning-based route selection [8], [9], [12]–[19]. These studies establish the importance of online path choice, but their results are difficult to compare because topology, allocator assumptions, feedback model, replay semantics, and disruption setting vary across papers. Our work addresses this gap with a matched evaluation grid that compares policy families under common topology, threat, allocator, and replay-capacity settings.

Bandit policies for routing decisions: Bandit methods provide a natural online-learning interface for quantum routing as each decision reveals partial feedback about the selected path, allocation, or context. We compare stochastic, contextual/neural, adversarial, predictive, and hybrid bandit policies rather than treating "online learning" as a single category. UCB and Thompson Sampling motivate stochastic baselines [20], [21]; EXP3 motivates adversarial baselines [22]; LinUCB, NeuralUCB, and NeuralTS motivate contextual and nonlinear reward models [23]–[26]; pursuit-style learning automata motivate structured exploitation [27]; and contextual multi-armed bandits (CMABs) and informed contextual multi-armed bandits (iCMABs) motivate context- and forecast-augmented policies [28], [29]. Huang *et al.* [10] employs *EXPNeuralUCB*, combining adversarial EXP3-style exploration with NeuralUCB-style reward modeling for joint path selection and qubit allocation. In contrast, we employ a stochastic approach; the key difference is evaluation scope: Huang *et al.* test a specific adversarial-neural policy, while our study compares multiple policy families under shared topology, threat regimes, allocator policies, and replay-capacity settings. This lets us attribute robustness differences to the policy–allocator–capacity interaction.

Benchmarking and matched robustness evaluation: Quantum-network benchmarking and utility-driven simulation frameworks motivate separating network behavior from policy behavior during evaluation [11], [30], while best-of-both-worlds bandit theory motivates testing across both stochastic and adversarial regimes [31]. Link-level and path-level best-arm formulations such as LinkSelFiE and related benchmarking approaches address unknown link or path qualities [9], [32], but they do not jointly test allocator policy, replay-capacity semantics, and threat progression. Our evaluation targets this matched-comparison gap: all bandit-policy families are tested under the same topology, threat regimes, allocator policies, and replay/capacity settings so that robustness differences can be attributed to the bandit-policy–allocator–capacity interaction rather than to incompatible experimental assumptions.

III. System Model

At each frame t, a routing agent selects path P_r and allocation $\mathbf{x}$ under uncertain link success and path availability. The model specifies the topology, reward process, threat regimes, allocators, and bandit interface used in evaluation.

A. Network Topology and Path Structure

4-node diamond topology: We use a four-node diamond topology with source S, destination D, and two intermediate repeaters, yielding four candidate paths $\mathcal{P} = \{P_1, P_2, P_3, P_4\}$: $P_1 = S \to B \to D$, $P_2 = S \to C \to D$, $P_3 = S \to B \to C \to D$, and $P_4 = S \to C \to B \to D$. We use r to index paths, so P_r is the selected path, while r' denotes a dummy path index for summations or comparisons. Each path P_r has hop count h_r, with $h_1 = h_2 = 2$ and $h_3 = h_4 = 3$; each hop denotes one adjacent entanglement link. This topology instantiates the path-length and routing-allocation tradeoffs introduced in Section II, while keeping the path set small enough for controlled matched-regime evaluation.

Qubit budget and allocator policies: The internal topology uses the baseline split $(T_1, T_2, T_3, T_4) = (8, 10, 8, 9)$, totaling 35 qubits; external testbeds use topology-specific budgets. We evaluate Fixed, ThompsonSampling, DynamicUCB, and Random allocators, spanning static, uncertainty-aware, optimistic, and uninformed allocation. Fixed maintains the baseline split; ThompsonSampling samples path utilities from posterior estimates; DynamicUCB redistributes capacity using upper-confidence estimates; and Random assigns capacity uniformly as a control. These allocators define the resource-management axis summarized in Table II. For path P_r with budget T_r, an allocation is $\mathbf{x} = (x_1, \ldots, x_{h_r})$ with $\sum_{\ell=1}^{h_r} x_\ell = T_r$, defining context space $\mathcal{X}_r$. For three-hop paths, this space grows combinatorially, motivating structure-aware approximation.

B. Reward Model and Link Success

Probabilistic entanglement generation: For path P_r with h_r links, let $p_e^{(\ell)} \in [10^{-4}, 2\times 10^{-4}]$ denote the per-attempt entanglement success probability of link ℓ, and let x_ℓ be the number of allocated qubits/attempts on that link during a frame. Assuming independent attempts within a frame, the link-level success probability is

$$p_\ell(x_\ell) = 1 - (1 - p_e^{(\ell)})^{x_\ell}.$$

Path-level success: Consistent with the multi-hop constraint summarized in Section II, path success requires all links on P_r to succeed [4], [6]. We define path success probability as

$$q_r(\mathbf{x}) = \prod_{\ell=1}^{h_r} p_\ell(x_\ell).$$

Thus, longer paths incur a multiplicative success penalty, while purification effects remain outside this routing model.

Bernoulli rewards under adversarial availability: At frame t, selecting path P_r with allocation $\mathbf{x}$ yields

$$Y_t(r, \mathbf{x}) \sim \mathrm{Bernoulli}\big(q_r(\mathbf{x}) A_t(r)\big).$$

Here $Y_t(r, \mathbf{x}) \in \{0, 1\}$ is the observed success indicator, and $A_t(r) \in \{0, 1\}$ is the availability gate, with $A_t(r) = 1$ when P_r is available and $A_t(r) = 0$ when disrupted. The threat regimes in Section III-C specify how $\mathbf{A}_t = (A_t(1), \ldots, A_t(4))$ evolves.

C. Adversarial Threat Taxonomy

We evaluate five threat regimes that specify the availability vector $\mathbf{A}_t$ introduced in Section III-B, forming a controlled escalation from no-disruption operation to online reactive disruption. Baseline separates routing and allocation from path unavailability; Stochastic captures independent random failures; Markov captures temporally correlated outages; and Adaptive/OnlineAdaptive test whether policies remain stable when disruption responds to recent routing behavior [22], [33], [34]. This progression helps attribute robustness differences to the policy–allocator–capacity interaction.

Concretely, Baseline keeps all paths available; Stochastic independently disrupts paths with fixed probability; Markov introduces temporally correlated on/off availability; Adaptive reacts to recent routing behavior over a sliding window; and OnlineAdaptive updates disruptions online using exponentially weighted path usage.

D. Multi-Armed Bandit Formulation

Joint action space: Using the topology and reward definitions above, each bandit action is a joint routing-allocation pair:

$$\mathcal{A} = \{(r, \mathbf{x}) : P_r \in \mathcal{P},\ \mathbf{x} \in \mathcal{X}_r\}.$$

At frame t, the policy selects $a_t = (r_t, \mathbf{x}_t) \in \mathcal{A}$ and observes only the realized reward $Y_t(r_t, \mathbf{x}_t)$ from Section III-B.

Policy interface: This interface instantiates the policy-family taxonomy introduced in Section II: bandit policies differ in how they estimate or sample action value, not in the underlying routing task. For contextual or neural policies, the context vector is derived from path identity, hop count, allocation vector, recent reward history, and threat/availability state when available. Non-contextual, adversarial, predictive, and hybrid variants consume the same action/reward stream but apply different scoring, exploration, or update rules.

The learner seeks to maximize cumulative expected success,

$$\max_{\pi}\ \mathbb{E}_\pi\left[\sum_{t=1}^{H} Y_t(r_t, \mathbf{x}_t)\right],$$

under the threat process and allocator semantics fixed for that evaluation run. We use this objective as the common empirical decision interface, not as a theoretical contribution.

E. Policy Implementation Interface

The evaluated policies share the joint action/reward interface in Section III-D but differ in how they score paths, allocate qubits, and update from feedback. Hybrid policies combine a path-level selection rule (e.g., adversarial weighting or pursuit) with an allocation-level contextual scorer.

Adversarial variants maintain path weights or probabilities over $r \in \{1, \ldots, |\mathcal{P}|\}$, then sample or select r_t before choosing an allocation. This connects the update mechanism to the adversarial-feedback setting in Section III-C. Given r_t, contextual or neural policies score feasible allocations $\mathbf{x} \in \mathcal{X}_{r_t}$ using reward estimates and uncertainty:

$$\mathbf{x}_t \in \arg\max_{\mathbf{x} \in \mathcal{X}_{r_t}} \left[\widehat{q}_{r_t}(\mathbf{x}) + \beta_t U_{r_t}(\mathbf{x})\right].$$

Informed variants augment the context vector from Section III-D with predictive features while leaving the environment, action space, and allocator interface unchanged.

IV. Study Design

A. Research Questions

Our work addresses three research questions:

- **RQ1.** *How do classical and context-aware multi-armed bandit (MAB) policies perform under stochastic quantum network conditions?* Stochastic decoherence alone separates viable from failing policies (Figure 3): `CPursuit`, `CEpsilonGreedy`, and `iCEpsilonGreedy` stay above 85%, while epoch-greedy designs collapse to ≈37%.
- **RQ2.** *How does the performance of bandit-based routing strategies change as network disruptions evolve from stochastic noise to structured, adaptive adversarial interference?* Context-aware representatives display stronger adversarial-scope stability (Figures 4 and 5): `CPursuit` and `iCEpsilonGreedy` reach 88.1% and 86.9% average efficiency with 77.4% and 81.0% floors, while `EXPNeuralUCB` falls to an 18.0% floor[2].
- **RQ3.** *How do choices in bandit policy, resource allocation strategy, and replay-capacity semantics interact to affect routing efficiency and stability in quantum entanglement routing?* Robustness is a joint function of algorithm, allocator, and capacity (Figures 6 and 7): under `ThompsonSampling` allocation and T_b-type anchoring, OnlineAdaptive efficiency drops by 4.4 pp from s=1 to s=1.5, then recovers by 6.0 pp at s=2.

B. Experimental Design

Section III defines the configuration axes to answer Section IV-A. Table I defines variables, and Table II summarizes design dimensions, tested options, and RQ coverage.

[2]Defined in Table I

TABLE I

NOTATION AND METRICS USED IN THE MATCHED EVALUATION.

Term	Meaning
`Oracle` gap	Difference between achieved efficiency and `Oracle` reference.
Floor	Lowest efficiency observed in the stated comparison scope; higher floors indicate stronger worst-case robustness.
CV	Coefficient of variation across scenario-level means; lower values indicate more stable performance.
s	Replay-capacity scale factor; larger s increases replay scale.
S	Number of independent runs for repeated stochastic evaluation.
F_b	Base frame horizon used for T_b-type replay anchoring.
F_c	Current frame horizon used for T-type replay anchoring.
T_b	Replay capacity anchored to base horizon: $T_b = sF_b$.
T	Replay capacity anchored to the current horizon: $T = sF_c$.

Network configuration: We use the four-path diamond topology shown in Figure 2 and defined in Section III-A, with two-hop paths P_1 and P_2 and three-hop paths P_3 and P_4. This topology keeps the routing action space nontrivial while preserving tractable cross-product sweeps across Table II.

Across all experiments, total physical budget remains fixed at 35 qubits, matching the resource constraint in Section III-A; this prevents over-provisioning and creates the intended exploration–exploitation pressure. Per-path budgets are set by allocator policies defined in Section III-A, enabling controlled comparison of static and adaptive resource management.

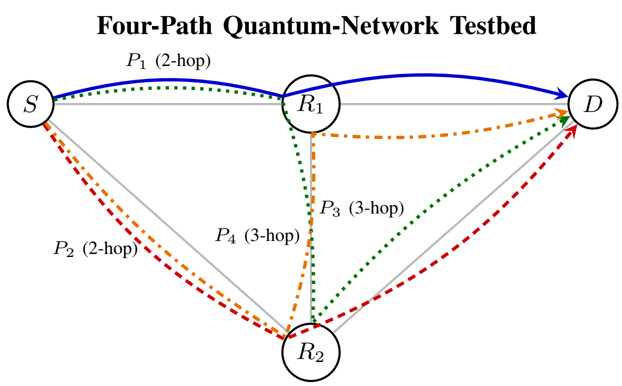


Fig. 2. The topology provides four candidate source–destination paths: two shorter two-hop paths (P_1, P_2) and two longer three-hop paths (P_3, P_4). This structure creates the path-length and qubit-allocation tradeoff evaluated throughout the study, with per-path budgets varied by allocator policy.

Time horizons: We evaluate `4K`, `6K`, and `8K`-frame horizons to capture short-, mid-, and long-episode learning dynamics. Primary results use `6K` unless noted; `4K`/`8K` comparisons support the RQ3 sample-efficiency analysis.

Replay configurations (capacity scaling): We consider base horizons $F_b \in \{4, 6, 8\}$K from the current horizon F_c used after frame scaling, and evaluate the number of independent runs (S), with $S \in \{3, 5, 8, 10\}$, per configuration. Replay memory is scaled by the replay-capacity scale factor (s), with $s \in \{1, 1.5, 2\}$ and default $s = 2$, under two semantics:

$$T_b = sF_b, \qquad T = sF_c.$$

TABLE II

EXPERIMENTAL DESIGN MATRIX LINKING CONFIGURATION DIMENSIONS, TESTED OPTIONS, AND RESEARCH-QUESTION COVERAGE.

Configuration Dimension	Options Tested	RQ(s)
Network topology	4-node, 4-path diamond	All
Time horizons	`4K`, `6K`, `8K` frames	RQ3
Qubit capacity	35 total qubits, fixed	All
Allocators	Fixed, ThompsonSampling, DynamicUCB, Random	RQ3
Replay capacity	$T_b = sF_b$, $T = sF_c$	RQ3
Threat models	Baseline, Stochastic, Markov, Adaptive, OnlineAdaptive	RQ1, RQ2
Forecasting	None, ARIMA ($n = 50$), ARIMA ($n = 100$)	RQ3
Bandit-policy families	Classical, Predictive, Adversarial, Contextual/Neural	RQ1, RQ2
Evaluation phases		
Ph 1	12 conditions (for MAB baseline)	RQ1
Ph 2	180 conditions per family (CMAB/iCMAB)	RQ2
Ph 3	240 conditions (for Dynamic allocation)	RQ3
Ph 4	120 conditions (for Capacity ablation)	RQ3

The s sweep, intermediate sensitivity checks, and doubled-current stress test probe the RQ3 capacity-paradox question: when added replay memory helps estimation versus when it increases adversarial predictability [35], [36].

Resource separation: Replay scaling changes only classical memory: the physical network budget is fixed at 35 qubits as defined in Section III-A and summarized in Table II, isolating replay-memory effects from quantum-network constraints.

Adversarial threat taxonomy: The five threat settings follow Section III-C and Table II; the scenario-specific parameters below instantiate the matched-interference comparison [10].

Scenario specifications: Table III instantiates the threat settings defined in Section III-C for the matched evaluation grid.

TABLE III

SCENARIO SPECIFICATIONS USED FOR MATCHED THREAT EVALUATION.

Scenario	Specification	RQ(s)
Baseline	0% disruption; benign-operation reference	RQ1
Stochastic	6.25% i.i.d. disruption [10]	RQ1-2
Markov	25% four-state structured disruption [10]	RQ2
Adaptive	25% high-usage, targeting over $w = 50$ frames [10]	RQ2
OnlineAdaptive	25% adaptive targeting with $\gamma = 0.97$ and softmax selection [10]	RQ2

Bandit-policy portfolio: The evaluated bandit-policy portfolio spans classical, adversarial, contextual/neural, pursuit-based, and predictive policies, with an Oracle reference enabling matched comparisons across policy families.

Key bandit-policy features: CPursuitNeuralUCB adds topology/channel-quality context to pursuit learning, while iCPursuitNeuralUCB adds ARIMA$(1, 0, 1)$ forecasts with warmup window size (n), using $n \in \{50, 100\}$ [29]. The Oracle normalizes efficiency to an ideal-reference baseline.

Qubit allocation strategies: Phase 3 varies the allocator policies defined in Section III-A and summarized in Tables II and IV while keeping the physical budget fixed at 35 qubits.

TABLE IV
QUBIT ALLOCATION STRATEGIES AND AVERAGE EFFICIENCY ACROSS SCENARIOS.

Allocator	Mechanism	Avg Eff.	Overhead
Fixed	Static distribution $(8, 10, 8, 9)$	87.7%	1.0×
Thompson	Bayesian posterior sampling [21], [37]	**88.2%**	0.3×
DynamicUCB	Adaptive UCB, $\lambda = 2.0$ [20]	85.3%	1.0×
Random	Uniform sampling, $\epsilon = 1.0$	77.3%	1.0×

Experimental scope and statistical protocol: The four phases in Table II progress from MAB baselines to contextual/adversarial bandit policies, allocator interactions, and replay-capacity ablations. Overall, the design covers 552 unique configurations with up to 10 runs each, targeting roughly 5,520 episodes and about 380 A100 GPU-hours. Runs use fixed seeds with repeated-run count $S \in \{3, 5, 8, 10\}$; primary results report 3- and 5-run averages.

We report Oracle-normalized efficiency,

$$\text{Eff}(\%) = \frac{\sum_{t=1}^{T} r_t}{T\theta^*} \times 100,$$

and coefficient of variation (CV),

$$\text{CV}(\%) = \frac{\sigma(\text{Eff})}{\mu(\text{Eff})} \times 100.$$

Significance uses 10,000-sample nonparametric bootstrap confidence intervals [38]; differences above 5 percentage points (pp) with non-overlapping intervals are practically significant, and head-to-head wins require a > 2 pp efficiency lead.

V. RESULTS

Utilizing the matched grid defined in Section IV, we report validated corpus results by Section IV-A: RQ1 tests stochastic viability, RQ2 tests threat escalation, and RQ3 tests allocator/capacity deployment effects. GA-report observations are used only as supporting context.

A. RQ1: Stochastic Routing Viability

Under the Stochastic scenario shown in Table III, contextual pursuit and epsilon-greedy baselines remain viable, while several context-free or poorly matched variants collapse (Figure 3). This establishes baseline viability before the structured and adaptive threats analyzed in RQ2.

Dataset and aggregation: RQ1 uses the EXP-family, CMAB, and iCMAB corpora, filtered to Stochastic runs under the default allocator. Results aggregate across 4K–8K horizons, replay-capacity scale factor (s) with $s \in \{1, 1.5, 2\}$, both current-horizon (T) and base-horizon (T_b) replay semantics, and the 3- and 5-run suites.

Table V shows a sharp stochastic-regime split: CPursuit and epsilon-greedy contextual variants remain at or above the 85% viability target, while CEXP4 and epoch-greedy variants

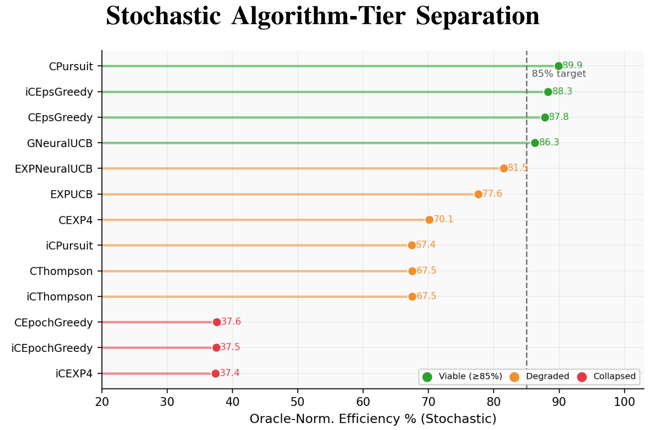


Fig. 3. Under stochastic disruption, `Oracle`-normalized efficiency separates viable contextual policies from failed designs. `CPursuit`, `CEpsGreedy`, and `iCEpsGreedy` remain robust, while epoch-greedy and structurally mismatched policies collapse.

TABLE V
RQ1 STOCHASTIC RESULTS FROM THE VALIDATED CORPORA. VALUES ARE MEAN ORACLE-NORMALIZED EFFICIENCY.

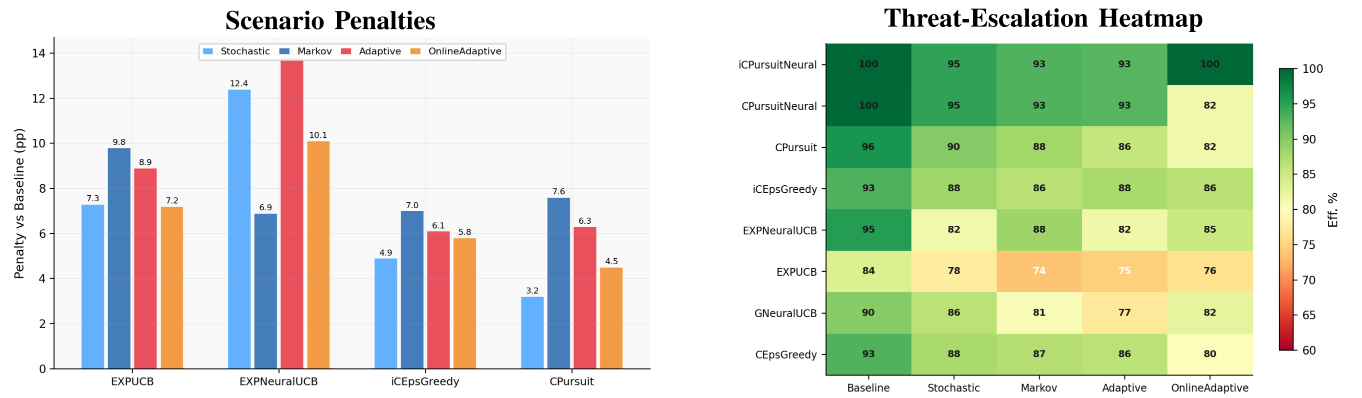
Model	3 runs	5 runs	Avg.
Top tier: viable under stochastic disruption			
CPursuit	89.3	90.6	**89.9**
CEpsilonGreedy	87.3	88.4	87.8
iCEpsilonGreedy	87.3	88.4	87.8
GNeuralUCB	83.7	87.4	85.5
Mid tier: degraded			
EXPNeuralUCB	81.6	79.6	80.6
EXPUCB	75.1	73.9	74.5
CThompsonSampling	65.5	67.9	66.7
iCPursuit	66.7	68.0	67.4
iCThompsonSampling	60.4	61.7	61.1
Collapsed: structural failure			
CEXP4	38.2	41.0	39.6
CEpochGreedy	35.6	38.3	37.0
iCEpochGreedy	35.6	38.3	37.0
iCEXP4	35.6	38.3	37.0

collapse to roughly 37–40%. The stochastic setting therefore separates viable contextual baselines from structural failures before capacity effects are disaggregated in RQ3.

B. RQ2: Robustness Under Adaptive Threats

Under Markov, Adaptive, and OnlineAdaptive threats, contextual and informed policies maintain stronger robustness floors than adversarial-first baselines (Figures 4 and 5). This extends RQ1 from stochastic viability to adaptive disruption.

Threat scope and aggregation: RQ2 uses the Markov, Adaptive, and OnlineAdaptive threat slice from Table III, restricted to the default allocator and aggregated across 4K–8K horizons, replay-capacity scale factor (s) with $s \in \{1, 1.5, 2\}$, both current-horizon (T) and base-horizon (T_b) replay semantics, and the 3- and 5-run suites. The table demonstrates the



(a) Compared with Baseline, `EXPNeuralUCB` incurs the largest Stochastic and Adaptive penalties (12.4 and 13.8 percentage points), while `CPursuit` has smaller penalties under Stochastic and OnlineAdaptive disruption (3.2 and 4.5 percentage points). Adversarial-first exploration does not guarantee robustness; contextual and pursuit-style policies preserve efficiency more consistently as threats escalate.

(b) Across Baseline, Stochastic, Markov, Adaptive, and OnlineAdaptive regimes, `iCPursuitNeural` preserves high efficiency (92.8% under Adaptive and 99.8% under OnlineAdaptive), while `EXPUCB` remains lower across Markov/Adaptive/OnlineAdaptive (73.8–76.4%), showing that adaptive threats expose stability gaps across model families.

Fig. 4. Panels (a) and (b) jointly show that contextual and informed models preserve stronger robustness as disruption progresses from stochastic noise to structured and adaptive interference.

strongest CMAB and iCMAB representatives along with EXP-family adversarial baselines.

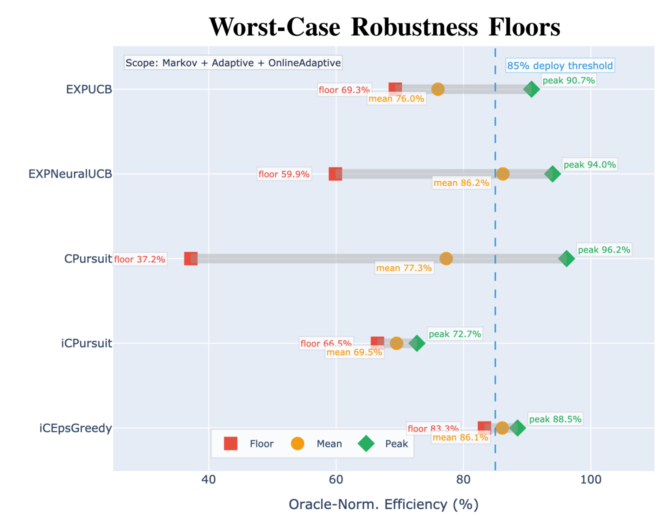


Fig. 5. The figure reports each representative policy's lowest `Oracle`-normalized efficiency across the locked Markov, Adaptive, and OnlineAdaptive scope. Higher floors indicate resistance to catastrophic collapse: `iCEpsGreedy` has the strongest floor, `CPursuit` has the best average, and `EXPNeuralUCB` remains fragile despite adversarial-style exploration. Together with Figure 4, this shows that adaptive threats expose stability gaps not visible from average efficiency alone.

Table VI and Figures 4 and 5 refute the adversarial-first hypothesis: CPursuit leads average efficiency, iCEpsilonGreedy provides the strongest stability floor, and EXPNeuralUCB remains fragile despite adversarial exploration. Structured and adaptive threats expose stability gaps not visible in RQ1, and adversarial-first EXP-family baselines do not dominate the locked adversarial scope.

TABLE VI
RQ2 ADVERSARIAL-SCOPE RESULTS. WIN DOMINANCE IS EACH DISPLAYED MODEL'S SHARE OF AGGREGATED SCENARIO WINS AMONG THE FOUR REPRESENTATIVES UNDER THE LOCKED MARKOV/ADAPTIVE/ONLINEADAPTIVE SCOPE.

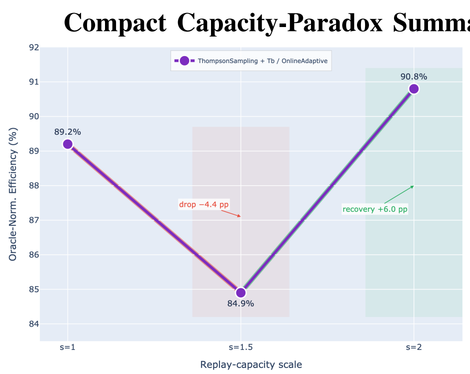
Algorithm	Avg.	CV	Floor	Win Dom.
CPursuit	**88.1**	5.3	77.4	25.6
iCEpsilonGreedy	86.9	**3.6**	**81.0**	**43.9**
EXPNeuralUCB	82.4	16.5	18.0	30.5
EXP3/UCB	76.3	6.0	68.8	0.0

C. RQ3: Deployment Optimization

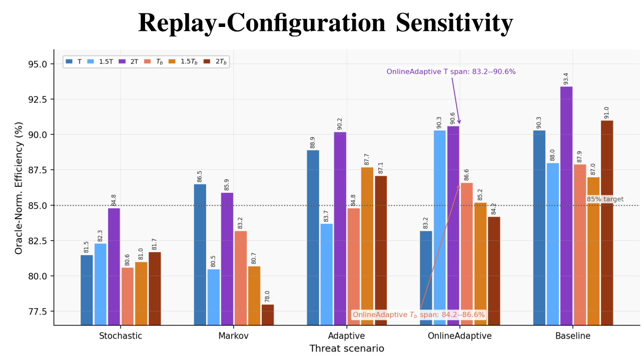
RQ3 evaluates whether algorithm, allocator, and replay-capacity choices should change with the threat regime. Deployment is configuration-sensitive, but the hybrid corpus also reveals a strong static configuration that exceeds the 85% robustness target across all five scenarios.

Deployment-evaluation corpus: RQ3 uses the validated Hybrid corpus for CPursuitNeuralUCB, iCPursuitNeuralUCB, GNeuralUCB, and EXPNeuralUCB across all scenarios, allocators, replay anchoring types, replay-capacity scales (s), with $s \in \{1, 1.5, 2\}$, and 4K–8K horizons. Unless otherwise noted, point estimates average the 3- and 5-run suites; RQ3 coefficient of variation (CV) values are computed across scenario means as a deployment-volatility proxy because per-run variance is not available in the corpus table.

Primary deployment observation: Deployment is configuration-sensitive, but the Hybrid corpus also reveals a strong static configuration. At the 6K horizon, `iCPursuitNeuralUCB+Fixed+(`T`-type,` $s = 2$`)` achieves a 96.0% global average and a 92.8% worst-case floor across scenarios, using the validated mean over 3- and



(a) Under the OnlineAdaptive threat, replay scaling is non-monotonic: efficiency drops from 89.2% at s=1 to 84.9% at s=1.5, then recovers to 90.8% at s=2. This pattern shows that added replay can help or hurt depending on the active threat.



(b) Across the six replay configurations, replay semantics changes the same scale effect: under OnlineAdaptive, T-type replay spans 83.2–90.6%, while T_b-type replay spans 84.2–86.6%. This shows that replay capacity must be interpreted jointly with anchoring rule and threat regime.

Fig. 6. Panels (a) and (b) jointly show that replay capacity acts as a threat-conditioned control variable governed by replay scale, anchoring rule, allocator behavior, and threat regime.

5-run suites. Scenario-specific optima still differ, so later RQ3 analyses disaggregate allocator and replay-capacity effects.

1) RQ3a: Predictive Context Modeling Impact: RQ3a isolates whether the informative/predictive variant, iCPursuitNeuralUCB, improves deployment robustness over reactive CPursuitNeuralUCB.

Controlled comparison: To isolate architecture impact, we fix deployment at 6K horizon, `Fixed` allocator, current-horizon (T)-type anchoring, and replay-capacity scale $s = 2$, using the validated 3-run suite from the Hybrid corpus.

TABLE VII
RQ3A INFORMATIVE-CONTEXT IMPACT UNDER THE FIXED DEPLOYMENT SETTING. VALUES ARE FROM THE VALIDATED 3-RUN HYBRID CORPUS.

Model	Bl	Sh	Mk	Ag	OA	Avg.	CV
CPursuitNeuralUCB	99.8	94.7	93.0	92.8	81.5	92.4	6.5
iCPursuitNeuralUCB	99.9	94.8	93.0	92.8	99.8	96.0	3.3

Bl=Baseline, Sh=Stochastic, Mk=Markov, Ag=Adaptive, OA=OnlineAdaptive; CV is computed across scenario means.

Table VII shows that iCPursuitNeuralUCB improves global robustness mainly by lifting OnlineAdaptive efficiency from 81.5% to 99.8% (+18.3 % points (pp)) while keeping Stochastic, Markov, and Adaptive performance unchanged. It reduces scenario-level dispersion from 6.5 to 3.3 CV. Thus, informative context is most valuable when the threat adapts online.

2) RQ3b: Replay Capacity Scaling & Paradox: RQ3b tests whether larger replay scale improves robustness or creates a threat-dependent capacity paradox.

Replay-scaling setup: Under `ThompsonSampling` allocation and base-horizon (T_b)-type replay anchoring, OnlineAdaptive efficiency drops by 4.35 pp from replay-capacity scale s=1 to s=1.5 and recovers by 5.96 pp at s=2. The fixed-`Random` slice in Table VIII is retained as supporting stress-control evidence: it shows that the same replay-sensitivity pattern becomes more severe under uninformed allocation, complementing the ThompsonSampling/T_b/OnlineAdaptive slice used for the paper-level RQ3b claim.

TABLE VIII
SUPPORTING RQ3B STRESS-CONTROL CAPACITY SCALING UNDER FIXED RANDOM ALLOCATION, 6K HORIZON, AND T-TYPE REPLAY ANCHORING. VALUES ARE HYBRID CORPUS MEANS OVER THE 3- AND 5-RUN SUITES.

Model	s	Bl	Sh	Mk	Ag	OA	Avg.
CPursuitNeuralUCB	1.0	94.7	93.8	89.3	88.5	81.5	89.6
	1.5	45.8	77.1	51.5	88.5	69.0	66.4
	2.0	96.3	94.9	95.9	88.5	84.5	92.0
iCPursuitNeuralUCB	1.0	92.1	93.6	88.7	88.5	70.8	86.7
	1.5	56.1	73.2	42.2	88.5	83.5	68.7
	2.0	95.1	94.5	96.3	88.5	83.5	91.6

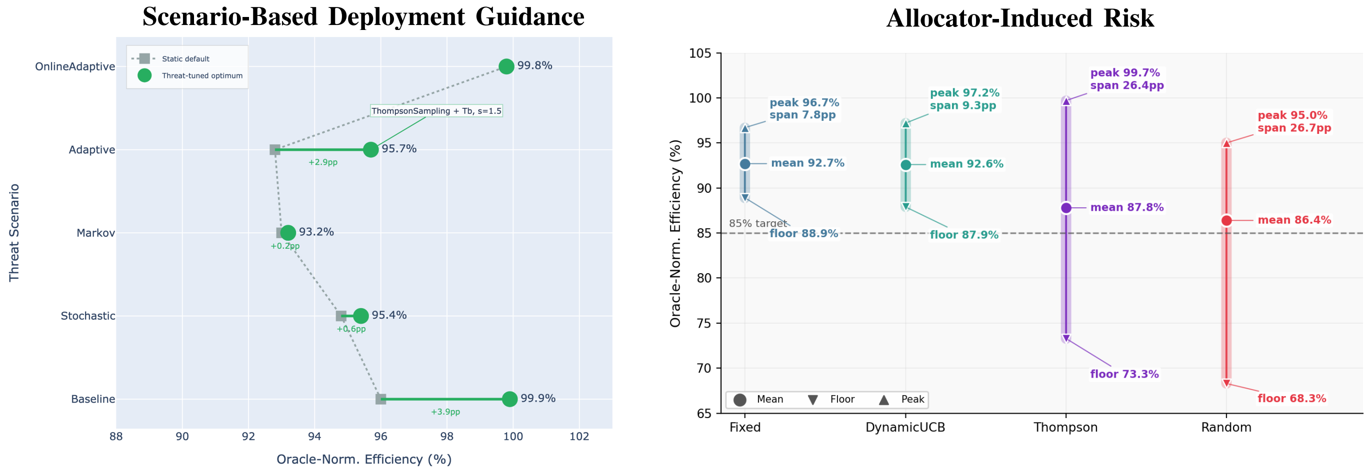
Bl=Baseline, Sh=Stochastic, Mk=Markov, Ag=Adaptive, OA=OnlineAdaptive; s is the replay-capacity scale factor.

Figure 6 and Table VIII show that replay scale acts as a threat-conditioned control variable. The allocator `ThompsonSampling`/T_b/OnlineAdaptive slice anchors the RQ3b conclusion; the fixed-`Random` table supports that conclusion as a stress-control comparison by showing sharper $s = 1.5$ degradation and $s = 2$ recovery under uninformed allocation. Replay capacity therefore must be tuned jointly with allocator and anchoring type.

3) RQ3c: Algorithm-Allocator Co-Design: RQ3c tests whether allocator choice is independent of algorithm architecture or instead acts as a first-class deployment decision.

Allocator-comparison setup: RQ3c fixes the model to iCPursuitNeuralUCB, the horizon to 6K, and capacity to current-horizon (T)-type replay with replay-capacity scale $s = 2$, then varies allocator policy. This isolates allocator effects after RQ3b isolates replay-scale behavior.

Table IX shows that allocator effects are deployment-critical: Fixed gives iCPursuitNeuralUCB the best average and floor, while Thompson is strong under Adaptive but fragile under Baseline/Stochastic mismatch. Thus, allocator selection is a first-class deployment decision.



(a) Comparing the baseline with scenario-specific switching, Adaptive disruption gives the largest validated gain: `ThompsonSampling` with T_b-type replay at s=1.5 reaches 95.7%, a 2.9 percentage-point improvement. This shows that deployment rules should match policy, allocator, and replay settings to the active threat.

(b) For `iCPursuitNeuralUCB`, allocator choice changes worst-case robustness: `Fixed (Default or Baseline)` keeps an 88.9% floor, while `Thompson` drops to 73.3%. This 15.6 percentage-point floor shift shows that allocator selection is a first-class deployment control.

Fig. 7. Panels (a) and (b) show that robust quantum-routing deployment is strongest when policy, allocator, and replay settings are matched to the threat.

TABLE IX
RQ3C ALLOCATOR INTERACTION FOR `ICPURSUITNEURALUCB` UNDER THE FIXED 6K, T-TYPE, $s = 2$ DEPLOYMENT. VALUES ARE HYBRID CORPUS MEANS OVER THE 3- AND 5-RUN SUITES.

Allocator	**Bl**	**Sh**	**Mk**	**Ag**	**OA**	**Avg.**	**Floor**
Fixed	99.9	94.6	94.7	88.5	99.8	95.5	88.5
DynamicUCB	85.0	89.5	88.9	88.5	97.1	89.8	85.0
Random	95.1	94.5	96.3	88.5	83.5	91.6	83.5
Thompson	58.4	65.4	87.8	91.7	65.2	73.7	58.4

Bl=Baseline, Sh=Stochastic, Mk=Markov, Ag=Adaptive, OA=OnlineAdaptive.

4) RQ3d: Scenario-Based Deployment Rules & Optimization: Deployment-rule synthesis: RQ3d converts the RQ3a–RQ3c findings into 6K deployment rules for iCPursuitNeuralUCB. Under the Hybrid corpus 3/5-suite mean used for switching analysis, the strong static default is `iCPursuitNeuralUCB+Fixed+`$(T$-type, $s = 2)$, with 95.5% average efficiency and an 88.5% floor across scenarios. Scenario-specific switching provides modest gains, with the largest validated improvement under Adaptive disruption.

TABLE X
RQ3D SCENARIO-SPECIFIC DEPLOYMENT RULES FOR `ICPURSUITNEURALUCB` AT 6K. GAIN IS MEASURED RELATIVE TO THE STRONG STATIC DEFAULT WHERE APPLICABLE.

Scenario	**Recommended configuration**	**Eff.**	**Gain**
Baseline	DynamicUCB + $(T, s = 1)$	99.9	+0.0
Stochastic	Thompson + $(T, s = 1)$	95.4	+0.6
Markov	DynamicUCB + $(T_b, s = 1.5)$	93.2	+0.3
Adaptive	Thompson + $(T_b, s = 1.5)$	95.7	+2.9
OnlineAdaptive	Fixed + $(T, s = 2)$	99.8	+0.0

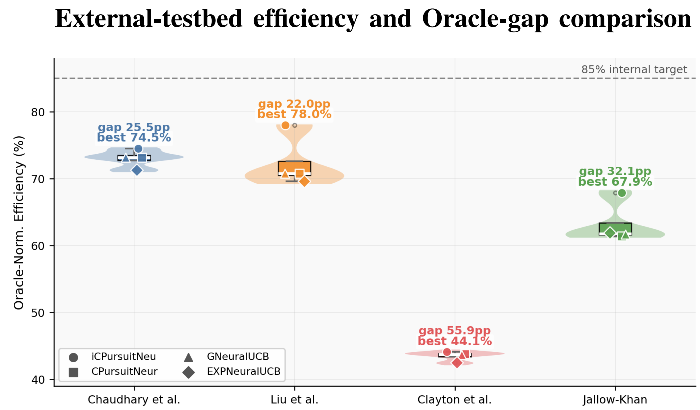
Table X and Figure 7 show the RQ3d conclusion: clear deployment rules emerge, but the static default remains strong enough to keep all scenarios above the 85% target at 6K. Threat-adaptive switching is most useful as a safeguard against the allocator/capacity mismatch diagnosed in RQ3b–RQ3c, with the largest validated gain under Adaptive disruption.

D. Cross-Testbed Validation

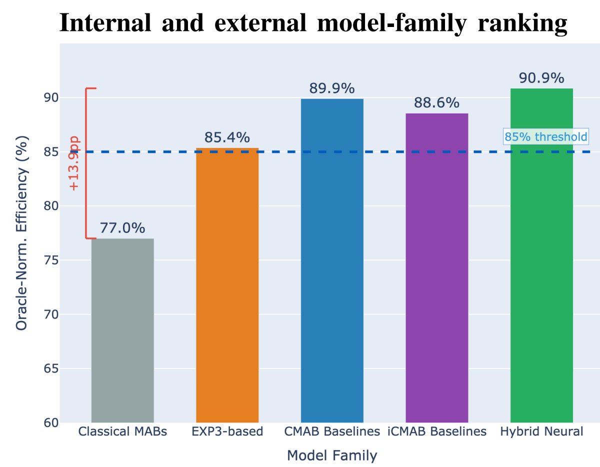
Figure 8 and Table XI show the cross-testbed claim in our work: pursuit–neural hybrids, especially `iCPursuitNeuralUCB`, provide the strongest average cross-testbed robustness, while high-complexity regimes can change the winner structure. Table XV gives the full per-configuration evidence supporting this hierarchy across the evaluated testbeds. Figure 8 and Table XII keep the model-family claim in the main body: hybrid pursuit–neural policies define the overall performance ceiling, and topology complexity reduces efficiency as routing spaces grow. The detailed family comparison remains available in Table XVI.

VI. DISCUSSION

Across corpus and external testbeds, pursuit-based hybrids consistently define the strongest efficiency–stability frontier, while adversarial-first baselines and poorly matched allocators expose lower robustness floors under structured and adaptive disruption. Our results show that routing robustness under compound uncertainty is jointly determined by *policy context*, *allocator choice*, and *replay-capacity semantics*. The appendix Table XVI shows `iCPursuitNeuralUCB` leading the internal model-family ranking with 90.9% average efficiency, ahead of `GNeuralUCB` (89.0%), and `EXPNeuralUCB` (88.4%), establishing pursuit mechanisms as the 1–3 pp differentiator within the neural-hybrid family. Similarly, Table XV shows that this aggregate advantage generalizes across external



(a) Across Chaudhary *et al.* [13], Liu *et al.* [12], Clayton *et al.* [16], and Jallow and Khan [15], `iCPursuitNeuralUCB` yields the strongest average cross-testbed efficiency, with leading averages ranging from 44.1% on Clayton *et al.* [16] to 78.0% on Liu *et al.* [12]. The larger `Oracle` gaps on harder topologies show that pursuit-informed neural routing transfers beyond the primary testbed but becomes harder to separate from the reference as topology complexity increases.



(b) All model families except classical MABs clear the 85% average-efficiency threshold, with hybrid neural policies highest at 90.9%. Across external testbeds, the same top-family pattern persists even as topology complexity compresses efficiency across models.

Fig. 8. Panels (a) and (b) jointly show that pursuit-informed neural designs generalize beyond the primary evaluation network, while increasing topology complexity widens `Oracle` gaps and compresses separation among models.

TABLE XI
CONCISE CROSS-TESTBED FINDINGS. FULL PER-ALGORITHM AND PER-TESTBED DETAIL IS REPORTED IN TABLE XV.

Testbed	**Claim-supporting finding**
Chaudhary *et al.* [13]	Hybrid models are tightly clustered, but `iCPursuitNeuralUCB` has the best average efficiency (74.5%) and most wins (95/300).
Liu *et al.* [12]	`iCPursuitNeuralUCB` dominates the dense 50-node topology, reaching 78.0% efficiency and 245/300 wins.
Clayton *et al.* [16]	All methods degrade on the 100-node fusion topology, but `iCPursuitNeuralUCB` leads with 44.1% efficiency and 97/300 wins.
Jallow and Khan [15]	`iCPursuitNeuralUCB` has the best average efficiency (67.9%), while `EXPNeuralUCB` has the largest win count (10/20), showing that average dominance and per-configuration dominance can diverge.

TABLE XII
CONCISE MODEL-FAMILY FINDINGS. FULL FAMILY-LEVEL DETAIL IS REPORTED IN TABLE XVI.

Scope	**Main finding**
Internal corpus	Hybrid pursuit–neural methods define the performance ceiling: `iCPursuitNeuralUCB` reaches 90.9%, above CPursuit (89.9%) and GNeuralUCB (85.4%).
External testbeds	`iCPursuitNeuralUCB` has the highest average efficiency on all four external testbeds, while Jallow and Khan [15] shows that EXP-style exploration can win more individual configurations.
Topology scaling	Efficiency declines as routing scale/complexity increases, from roughly 90% internally to 74–78% on Chaudhary *et al.* [13]/Liu *et al.* [12] and about 42–44% on Clayton *et al.* [16].

testbeds: `iCPursuitNeuralUCB` achieves the highest average efficiency on Chaudhary *et al.* [13] (74.5%), Liu *et al.* [12] (78.0%), Clayton *et al.* [16] (44.1%), and Jallow and Khan [15] (67.9%). While `EXPNeuralUCB` wins more configurations on Jallow and Khan [15], `iCPursuitNeuralUCB` maintains higher average efficiency (67.9% vs. 61.9%), showing more consistent performance across allocator and capacity settings. Both tables support the main hierarchy: pursuit-informed neural designs provide the strongest average robustness. Figures 10 and 11 provide the corresponding visual audit: the first figure connects Oracle-gap, qubit-budget, and policy-ranking diagnostics, while the second groups replay drop/recovery, allocator, context-capacity, capacity-trajectory, and regret evidence behind the deployment interpretation.

Context-Aware Bandit Policies vs the Robustness Frontier:

Across the paper baselines (RQ1–RQ2) and the Hybrid suite (RQ3), context-aware policies outperform non-contextual and purely adversarial baselines in the settings that matter most for deployment. Pursuit-based contextual hybrids (`CPursuitNeuralUCB`, `iCPursuitNeuralUCB`) maintain strong efficiency across the five threat regimes, while EXP3-family baselines show weaker robustness floors under Markov, Adaptive, and OnlineAdaptive disruption. This suggests that topology and channel-state context are core routing features needed for robust quantum-network operation.

The Capacity Paradox is Conditional:

Increasing replay capacity can improve stability in structured regimes while amplifying vulnerability under reactive threats. The effect emerges from the interaction between replay capacity and allocator-driven exploration predictability: larger replay buffers can reinforce recurring patterns, and reactive adversaries can exploit those patterns. Capacity sensitivity is most severe under allocator stress, particularly with `Random`. In the Hybrid suite, efficiency swings exceed 20–38 percentage points across replay-capacity settings under Adaptive and Markov threats. Replay capacity must therefore be tuned jointly with allocator policy and threat regime.

Allocator Choice is the Practical Deployment Lever:

Allocator choice is the highest-leverage deployment control in the Hybrid suite. Aggregated across scenarios and replay-capacity settings, `ThompsonSampling` and `Fixed`/`Default` form the top performance tier, while `Random` is strongly associated with catastrophic efficiency collapses. Allocator choice therefore functions as a safety mechanism and a first-class deployment control. Even strong pursuit-neural architectures can be pulled into the capacity paradox under allocator mismatch, while robust allocators substantially suppress replay sensitivity.

A. Implications for Adversarial Quantum Network Design

The results imply three design principles for adversarial quantum-network routing: context-aware routing should be treated as baseline infrastructure, replay capacity should be tuned as a coupled control variable, and deployment settings should adapt to the active threat regime. First, feature enrichment through topology, link-state, and traffic information yields larger robustness gains than refining reward-only exploration rules. Second, replay capacity is a coupled deployment decision: the anchoring rule (T_b vs. T), scale (s), and allocator jointly determine tail risk under structured and adaptive disruption. Finally, lightweight threat-aware meta-policies should use `ThompsonSampling` or the static `Default` allocator as conservative deployment choices, treat replay-capacity settings as scenario-coupled, and adapt allocator and replay configurations using observed efficiency and variability signals.

VII. Limitations and Future Work

A. Limitations

The main limitations of this study are *topology scale*, *testbed fidelity*, *replay-capacity* coverage, and *forecasting scope*. While these constraints enable rigorous controlled experiments, they bound the generality of results and motivate validation across larger, more heterogeneous quantum-network testbeds. *First*, all experiments use a controlled 4-node, 4-path internal topology with mid-range horizons, which enables clean attribution of algorithm, allocator, and replay-capacity effects but may not capture behavior in larger or more heterogeneous quantum-network testbeds. *Second*, the current software-defined testbeds abstract away control-plane latency, node-level scheduling delays, memory contention, and timing-dependent decoherence effects, so deployment-level conclusions require further validation across higher-fidelity and more diverse testbeds. *Third*, replay capacity is evaluated using two anchoring rules (`Tb`, `T`) and three scale factors ($s \in \{1, 1.5, 2\}$), which exposes the `capacity paradox` and motivates broader tail-risk characterization at larger horizons and topologies. *Lastly*, predictive context currently uses ARIMA-based forecasting; more expressive temporal models (*e.g.,* RNNs and Transformers) with calibrated uncertainty quantification remain future work.

B. Future Work

Future work extends the limitations found along four directions: *expanded testbed stress testing*, *cross-testbed standardization*, *larger routing and allocation spaces*, and *automated threat-aware adaptation*. *First*, we will evaluate pursuit–neural and other hybrid models on larger quantum-network testbeds under operational stressors—control-plane latency, node-level scheduling delays, memory contention, and timing-dependent decoherence—to stress-test robustness claims in higher-fidelity testbed settings. *Second*, we will conduct standardized cross-testbed benchmarking across sparse, compact, dense, and mid-scale topologies using consistent run configurations, determining whether pursuit–neural robustness and the capacity paradox generalize across network structures. *Third*, we will extend the framework to larger routing graphs, expanded action spaces, and hierarchical allocation strategies, studying allocator–capacity interactions under increasingly combinatorial conditions. *Lastly*, we will replace the current heuristic meta-policy with a learned threat detector and automated configuration selector that dynamically adapts allocator and replay-capacity settings under changing attack patterns.

VIII. Conclusion

This work shows that robust quantum entanglement routing under adversarial conditions requires joint control of path selection, qubit allocation, replay capacity, and threat response. The matched evaluation identifies `iCPursuitNeuralUCB` as the strongest context-aware neural routing model: it reaches 90.9% internal average efficiency, clears the 90% model-family threshold, and outperforms the closest internal alternatives, which remain below that mark. Under harder threats, adversarial-first exploration remains less reliable: `EXPNeuralUCB` falls to an 18% robustness floor, while the strongest contextual baselines maintain floors near 77–81%. The capacity paradox shows why deployment tuning matters: replay capacity must be matched to the active threat, and allocator choice directly shapes routing robustness. When the threat is unknown, a fixed pursuit–neural deployment baseline provides a strong starting point; when the threat is detected, allocator and replay settings should be adapted to the threat regime. The external-testbed comparison shows why the proposed approach is stronger than a topology-specific result: `iCPursuitNeuralUCB` leads average efficiency on Chaudhary *et al.* [13] (74.5%, 95/300 wins), Liu *et al.* [12] (78%, 245/300 wins), Clayton *et al.* [16] (44%, 97/300 wins), and Jallow and Khan [15] (67.9%). The advantage is clearest on the dense 50-node Liu *et al.* [12] setting, where it reaches 78% versus 70.8% for the next strongest models, and it still remains best on the larger Clayton *et al.* [16] fusion topology, where methods compress tightly at 44% versus 43.8%.

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Appendix A
System Model Layout

Figure 9 and Tables XIII and XIV preserve the setup artifacts behind the validated experiments: the control loop, tuned hyperparameters, and evaluated policy portfolio.

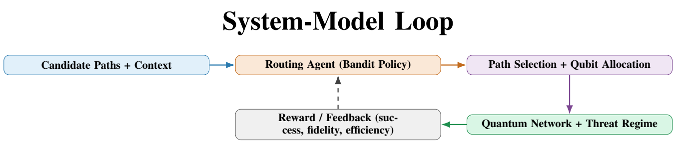


Fig. 9. At frame t, the routing controller observes candidate-path context, jointly selects a path and qubit allocation, interacts with the threat-conditioned quantum-network testbed, and updates from reward feedback. The loop supports the paper's core modeling claim: routing robustness must be evaluated as a coupled path-selection, allocation, threat, and feedback process.

TABLE XIII
Hyperparameter settings and literature justifications.

Parameter	Algorithm	Value	Ref.
Neural LR	EXPNeuralUCB	0.01	[10]
	GNeuralUCB	0.2	[25]
Exp. Weight	EXPNeuralUCB	0.05	[10]
	EXPUCB	0.005	Tuned
Pursuit LR	Pursuit family	0.2	[27]
UCB Explore	DynamicUCB	2.0	[20]
ARIMA (n)	iCPursuit	50, 100	[29]
Replay semantics	All	$T_b = sF_b$, $T = sF_c$	[35], [36]

Scenario specifications: Table III instantiates the threat settings defined in Section III-C for the matched evaluation grid.

TABLE XIV
Bandit-policy portfolio by evaluation phase.

Phase	Category	Bandit policies / reference
Phase 1	Classical MAB	LinUCB [23], LinTS [37], UCB1 [20], Thompson Sampling [21]
Phase 1–2	Adversarial	EXP3 [22], EXPUCB [10], EXPNeuralUCB [10]
Phase 2	Contextual/Neural	GNeuralUCB, NeuralUCB [25], NeuralTS [26], CEpsilonGreedy, CThompsonSampling
Phase 2–3	Pursuit-based	CPursuitNeuralUCB
Phase 3	Predictive	iCPursuitNeuralUCB
Baseline	Oracle	Perfect-information reference

Appendix B
Detailed Cross-Testbed and Model-Family Results

This appendix reports the full cross-testbed and model-family evidence supporting the concise Results summaries in Tables XI and XII. The tables preserve the per-testbed, per-family, and per-configuration detail behind the robustness hierarchy discussed in the main text.

A. Full Cross-Testbed Results

Table XV expands the concise main-body summary in Table XI. It shows the configuration-level evidence behind the cross-testbed claim that pursuit–neural hybrids, especially `iCPursuitNeuralUCB`, provide the strongest average robustness while higher-complexity regimes can alter the winner structure.

B. Full Model-Family Results

Table XVI expands the model-family summary from Table XII. It provides the per-family evidence behind the claim that contextual and pursuit–neural methods form the strongest efficiency–stability frontier, while EXP-family and structurally mismatched baselines retain weaker robustness floors.

Appendix C
Additional Diagnostic Figures

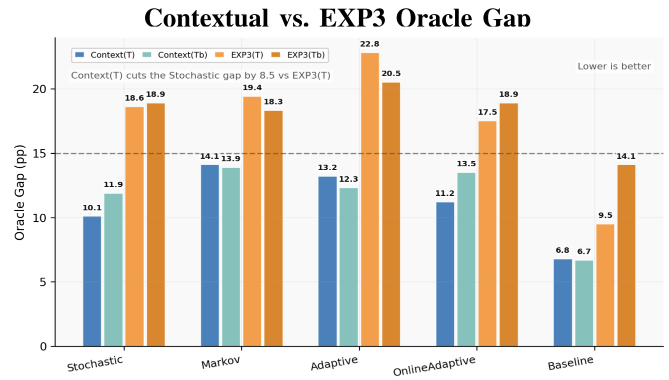
This appendix collects diagnostic figures that complement the main Results figures without duplicating plots already used in the paper body. The main paper keeps the claim-level figures, while the appendix preserves additional evidence for readers who want broader scenario, capacity, allocator, and cross-testbed views.

A. Additional Claim Support

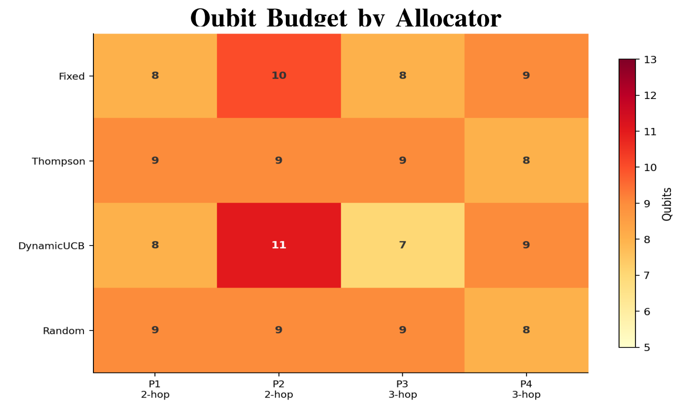
The following figures provide supporting views for claims that are central to the paper but too detailed for the main Results flow. They extend the main-body evidence in Figures 6 to 8: Figure 10 pairs Oracle-gap, budget, and policy-selection evidence on scenario efficiency, cross-threat gaps, mean Oracle gap, and overall ranking, while Figure 11 groups model, replay, allocator, capacity, and regret views into one appendix synthesis.

B. Grouped Synthesis Diagnostics

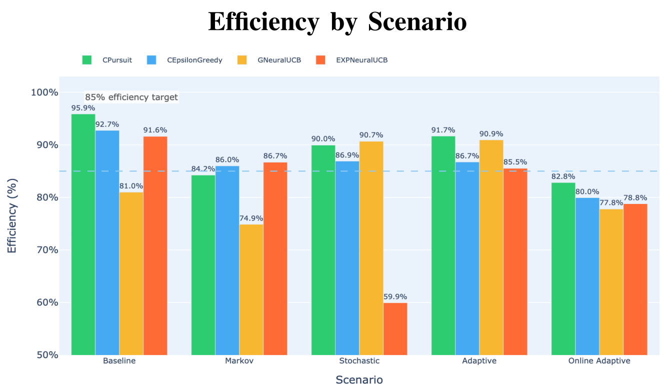
The grouped synthesis figure below remains in the appendix because it lays out the diagnostic panels as a wider six-panel synthesis. Figure 11 serves as the appendix-side reference point when readers want one view of how model-family gaps, replay drop/recovery behavior, allocator efficiency, context-capacity behavior, capacity-paradox trajectories, and regret views connect.



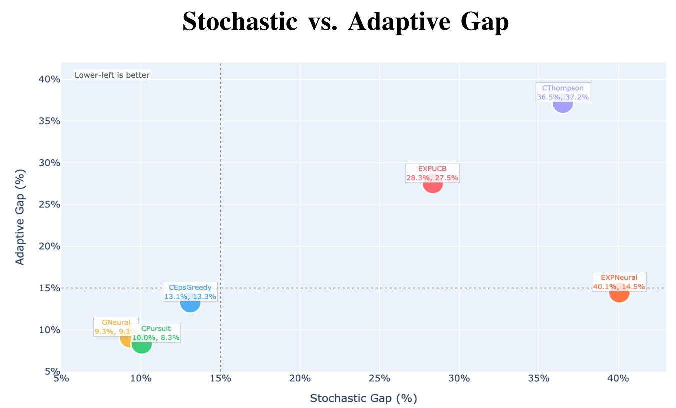
(a) Contextual policies show lower percentage-point distance from `Oracle` than EXP3-style baselines across scenario and capacity settings, meaning path/context information reduces efficiency loss under escalating disruption.



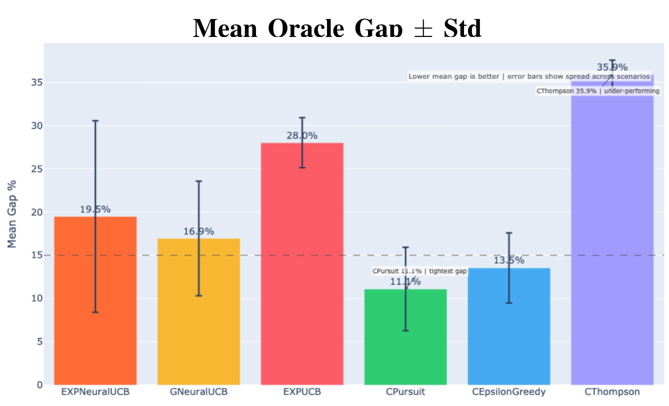
(b) The heatmap shows the 35-qubit internal split across paths and allocators, including the `Fixed` $(8, 10, 8, 9)$ baseline and allocator-driven path shifts; resource allocation is part of the evaluated routing action.



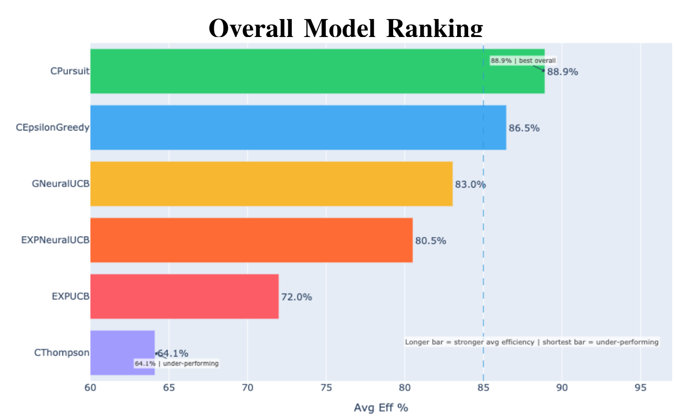
(c) `CPursuit` remains strongest across most scenario-level efficiency regimes, while `EXPNeuralUCB` drops sharply under Stochastic; policy selection separates robust and fragile regimes before replay-capacity tuning is applied.



(d) Models close to the Stochastic-vs-Adaptive origin stay near `Oracle` in both regimes, while separated points expose threat-specific weakness; one policy can tolerate a capacity change in one regime and fail in another.

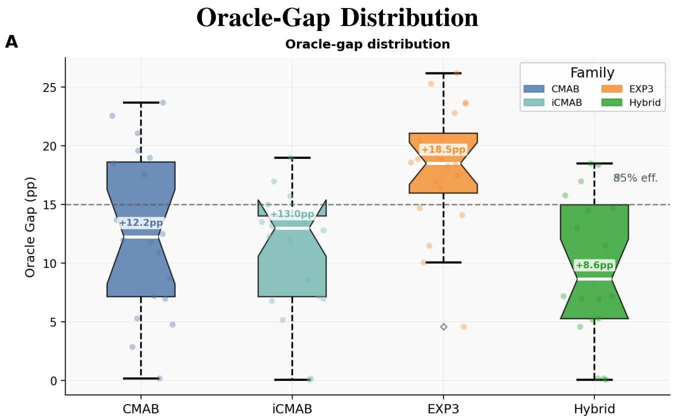


(e) Pursuit/contextual families hold lower mean distance to `Oracle` across scenarios, while `CThompson` and `EXPUCB` remain higher-gap options; the spread shows the stable policy margin available before replay-capacity tuning.

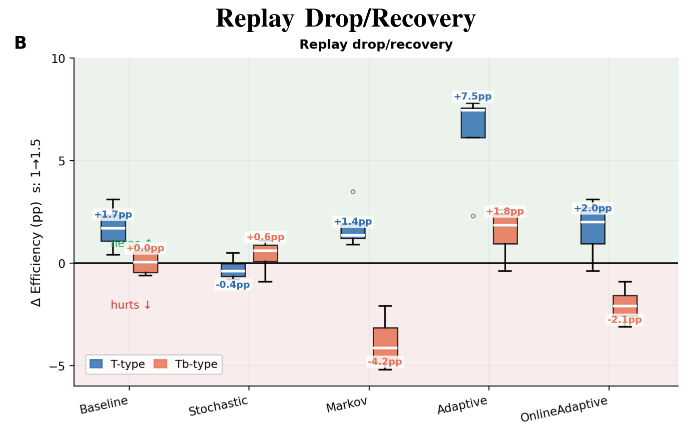


(f) `CPursuit` defines the strongest average-efficiency frontier and `CThompson` the weakest; capacity choices modulate but do not erase family-level robustness differences.

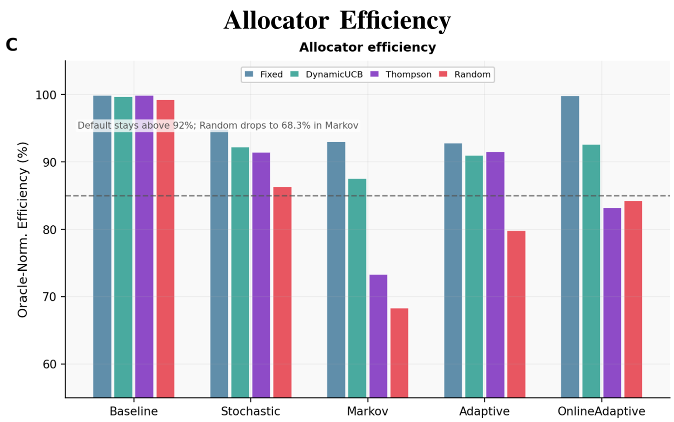
Fig. 10. Panels (a)–(f) show the appendix policy-selection evidence chain: Oracle-gap and qubit-budget context in (a)–(b) frame the routing-resource setting, while scenario efficiency, cross-threat Oracle-gap shape, mean-gap stability, and overall ranking in (c)–(f) show that replay-capacity effects must be read against the underlying policy frontier rather than in isolation.



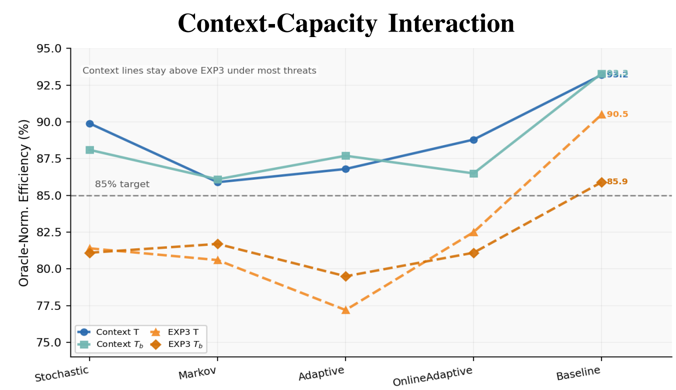
(a) EXP3 has the largest Oracle-gap losses, while hybrid/contextual families concentrate closer to the 15 pp line; context-aware and pursuit–neural designs define the stability frontier.



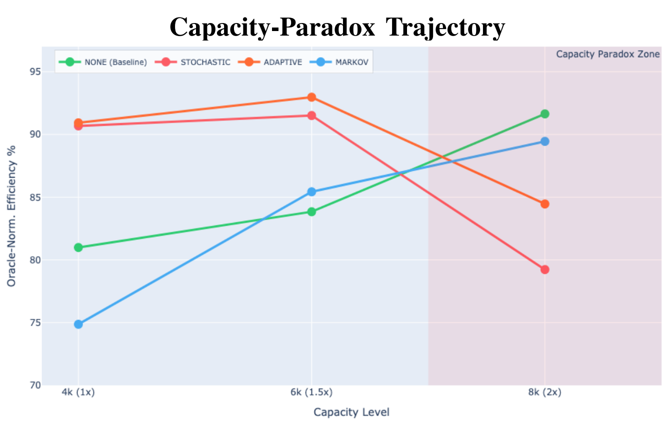
(b) Replay effects change sign across threats and replay semantics, with Markov and OnlineAdaptive showing the clearest helpful-to-harmful flip behind the capacity paradox.



(c) `Fixed` remains strongest for `iCPursuitNeuralUCB` across allocator-efficiency scenarios, while `Random` and mismatched adaptive allocators expose lower floors and allocator-driven tail risk.



(d) Contextual T/T_b curves stay near or above 85% across family/threat capacity settings, while EXP3 variants remain lower until Baseline; replay robustness depends on path/context structure.



(e) Stochastic and Adaptive rise from 4K to 6K and then fall at 8K along the capacity-paradox trajectory, while Markov continues upward; more capacity is not uniformly beneficial.

(f) `CThompsonSampling` and `EXPUCB` accumulate larger normalized regret over capacity steps than pursuit/contextual baselines, showing the learning-cost side of capacity-sensitive routing.

Fig. 11. Together, the panels show a grouped appendix synthesis: model-family Oracle gaps define the policy hierarchy, replay drop/recovery reveals threat-conditioned sign flips, allocator efficiency and context-capacity behavior expose complementary control-surface effects, and the capacity-paradox and regret views extend the figure with the broader replay/capacity story.

TABLE XV

FULL CROSS-TESTBED EVIDENCE FOR THE ROBUSTNESS HIERARCHY. THE TABLE REPORTS SCENARIO-AGGREGATED ORACLE-NORMALIZED EFFICIENCY, ORACLE GAP, REGRET, AND CONFIGURATION-LEVEL WINS FOR EACH EXTERNAL TESTBED. ACROSS THE FOUR TESTBEDS, iCPURSUITNEURALUCB IS THE STRONGEST AVERAGE-EFFICIENCY PERFORMER, WHILE THE WIN COUNTS SHOW THAT HIGHER-COMPLEXITY REGIMES CAN CHANGE WHICH LEARNER WINS INDIVIDUAL CONFIGURATIONS. TOGETHER, THE ROWS SUPPORT THE CROSS-TESTBED CLAIM THAT PURSUIT–NEURAL HYBRIDS PROVIDE THE MOST RELIABLE AVERAGE ROBUSTNESS, WHILE TOPOLOGY SCALE AND TESTBED CONSTRAINTS REDUCE SEPARATION FROM THE ORACLE. NUMERIC COLUMNS ARE MEANS ACROSS ALL FIVE SCENARIOS AND FOUR ALLOCATORS. CHAUDHARY *et al.* [13], LIU *et al.* [12], AND CLAYTON *et al.* [16] USE THE FULL EVALUATION CORPORA AGGREGATED ACROSS $s \in \{1.0, 1.5, 2.0\}$ (CAP_TYPE=T) WITH FIVE RUNS PER SETTING; JALLOW AND KHAN [15] USES THE CURRENT ORIGINAL PAPER-CONFIG SLICE AT 1K/1K/1R, $s = 1.0$, CAP_TYPE=T.

Testbed	Algorithm	Avg Reward	Regret	Eff.	Gap	Wins
Chaudhary *et al.* [13] 15N/51E/8P 4K/2K/5R 4 allocs, $s \in \{1, 1.5, 2\}$	`ORACLE`	0.4	0.0	—	—	—
	`CPursuitNeuralUCB`	0.3	509.2	73.2	26.8	38/300
	`GNeuralUCB`	0.3	515.9	73.2	26.8	84/300
	`iCPursuitNeuralUCB`	0.3	494.5	**74.5**	25.6	**95/300***
	`EXPNeuralUCB`	0.3	583.2	71.3	28.8	83/300
	Takeaway	All pursuit hybrids remain competitive; iCPursuitNeuralUCB has the highest average efficiency and most wins.				
Liu *et al.* [12] 50N/141E/15P 50/50/5R 4 allocs, $s \in \{1, 1.5, 2\}$	`ORACLE`	8.9	0.0	—	—	—
	`iCPursuitNeuralUCB`	7.0	42.8	**78.0**	22.0	**245/300***
	`EXPNeuralUCB`	6.2	272.3	69.6	30.4	16/300
	`CPursuitNeuralUCB`	6.3	269.1	70.8	29.2	0/300
	`GNeuralUCB`	6.3	269.1	70.8	29.2	39/300
	Takeaway	iCPursuitNeuralUCB dominates the dense topology in both efficiency and configuration-level wins.				
Clayton *et al.* [16] 100N/426E/4P 1.5K/500/5R 4 allocs, $s \in \{1, 1.5, 2\}$	`ORACLE`	0.9	0.0	—	—	—
	`GNeuralUCB`	0.4	864.1	43.7	56.3	53/300
	`EXPNeuralUCB`	0.4	892.2	42.5	57.5	91/300
	`iCPursuitNeuralUCB`	0.4	858.3	**44.1**	55.9	**97/300***
	`CPursuitNeuralUCB`	0.4	861.2	43.8	56.2	59/300
	Takeaway	The mid-scale fusion topology compresses all methods, but iCPursuitNeuralUCB still leads.				
Jallow and Khan [15] 20N/19E/8P 1K/1K/1R 4 allocs, $s = 1$	`ORACLE`	300540.2	0.0	—	—	—
	`iCPursuitNeuralUCB`	210419.0	60833.4	**67.9**	32.2	1/20
	`EXPNeuralUCB`	186656.1	82175.2	61.9	38.1	**10/20***
	`GNeuralUCB`	183038.8	83361.0	61.7	38.4	4/20
	`CPursuitNeuralUCB`	182011.6	84024.2	61.4	38.6	5/20
	Takeaway	iCPursuitNeuralUCB leads average efficiency, while EXPNeuralUCB wins more individual configurations.				

Legend: N=nodes, E=edges, P=paths, R=runs. Efficiency (%) = model average reward divided by Oracle average reward. Gap = 100 - Efficiency. Wins denote experiment-level wins out of the available configurations: 300 for Chaudhary *et al.* [13]/Liu *et al.* [12]/Clayton *et al.* [16] and 20 for Jallow and Khan [15]. * indicates the testbed-level experiment-winner count.

TABLE XVI

FULL MODEL-FAMILY EVIDENCE FOR THE EFFICIENCY–STABILITY HIERARCHY. THE INTERNAL BLOCKS COMPARE CMABS, ICMABS, EXP3-BASED METHODS, AND HYBRID NEURAL POLICIES, WHILE THE EXTERNAL BLOCKS REPORT PURSUIT–NEURAL PERFORMANCE ACROSS FOUR TESTBEDS. THE TABLE SHOWS THAT CONTEXTUAL AND INFORMED CONTEXTUAL BASELINES PROVIDE STRONG INTERNAL FLOORS, EXP-FAMILY METHODS REMAIN WEAKER IN TAIL BEHAVIOR, AND `iCPursuitNeuralUCB` GIVES THE HIGHEST HYBRID-NEURAL CEILING AND STRONGEST AVERAGE EXTERNAL PERFORMANCE. TOGETHER, THE ROWS SUPPORT THE PAPER'S CLAIM THAT ROBUST QUANTUM ROUTING DEPENDS ON MODEL FAMILY AND CONTEXT/PURSUIT STRUCTURE, NOT ONLY ON A SINGLE LEARNER SCORE. BEST PER-FAMILY/TESTBED PERFORMANCE IS BOLDED.

Scope	Algorithm	Avg Eff.	Gap	Floor	Wins
CMABs internal corpus	`CPursuit`	**89.9**	10.1	77.4	**54/75***
	`CEpsilonGreedy`	88.1	11.9	79.4	21/75
	`CEXP4`	70.1	29.9	67.4	0/75
	`CThompsonSampling`	68.2	31.8	62.5	0/75
	`CEpochGreedy`	37.7	62.4	36.3	0/75
	Takeaway	CPursuit is the strongest classical contextual baseline.			
iCMABs internal corpus	`iCEpsilonGreedy`	**88.6**	11.4	81.0	**75/75***
	`iCPursuit`	68.7	31.3	56.9	0/75
	`iCThompsonSampling`	68.0	32.0	62.8	0/75
	`iCEXP4`	37.5	62.5	36.1	0/75
	`iCEpochGreedy`	37.6	62.4	36.1	0/75
	Takeaway	iCEpsilonGreedy provides the strongest iCMAB floor and complete win dominance in this slice.			
EXP3-based internal corpus	`GNeuralUCB`	**85.4**	14.6	61.6	**41/75***
	`EXPNeuralUCB`	80.9	19.1	18.0	28/75
	`EXPUCB`	78.1	21.9	68.8	6/75
	Takeaway	GNeuralUCB is the strongest EXP-family representative, but its floor remains weaker than the top contextual variants.			
Hybrid neural internal corpus	`iCPursuitNeuralUCB`	**90.9**	9.1	76.4	**23/75***
	`CPursuitNeuralUCB`	89.0	11.0	45.0	17/75
	`GNeuralUCB`	89.0	11.0	62.2	21/75
	`EXPNeuralUCB`	88.4	11.6	63.9	14/75
	Takeaway	Hybrid pursuit-neural policies define the strongest internal performance ceiling.			
Chaudhary *et al.* [13] external	`iCPursuitNeuralUCB`	**74.4**	25.6	54.4	**25/75***
	`CPursuitNeuralUCB`	73.2	26.9	48.3	8/75
	`GNeuralUCB`	73.2	26.8	46.7	20/75
	`EXPNeuralUCB`	71.3	28.7	41.6	22/75
	Takeaway	iCPursuitNeuralUCB is the best average and win-count performer.			
Liu *et al.* [12] external	`iCPursuitNeuralUCB`	**77.5**	22.5	28.7	**58/75***
	`GNeuralUCB`	70.9	29.1	41.5	10/75
	`CPursuitNeuralUCB`	70.9	29.1	41.5	0/75
	`EXPNeuralUCB`	69.5	30.5	33.0	7/75
	Takeaway	iCPursuitNeuralUCB dominates the dense 50-node external setting.			
Clayton *et al.* [16] external	`iCPursuitNeuralUCB`	**41.6**	58.5	23.9	**26/75***
	`CPursuitNeuralUCB`	41.2	58.8	22.6	13/75
	`GNeuralUCB`	41.1	58.9	22.8	14/75
	`EXPNeuralUCB`	40.2	59.8	21.7	22/75
	Takeaway	All methods degrade under the larger fusion topology, but iCPursuitNeuralUCB remains best.			
Jallow and Khan [15] external	`iCPursuitNeuralUCB`	**67.4**	32.6	**44.1**	1/5
	`EXPNeuralUCB`	62.1	37.9	23.9	**2/5***
	`GNeuralUCB`	61.8	38.2	36.0	0/5
	`CPursuitNeuralUCB`	61.7	38.4	35.2	**2/5***
	Takeaway	iCPursuitNeuralUCB leads average efficiency; EXP/CP pursuit variants split the small-slice win count.			

Note: Internal corpus rows use the validated CMAB, iCMAB, EXP3, and Hybrid master datasets with `Default` allocator and reported run/scale settings. External rows use the standardized external-testbed corpus where available. Gap = 100 - Efficiency. Floor = worst-case efficiency. Wins denote experiment-level win counts; * marks the row with the highest win count in its scope.